



Data Article

SpiceSpectrum: Class-balanced dataset of commercially valuable spice cultivars



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ABSTRACT

This data paper introduces a comprehensive visual dataset of 11 globally significant spices, comprising 11,000 high-quality images curated to support research and applications in agriculture, trade, and culinary arts. The dataset features the following spices: Black Pepper, Cardamom, Cinnamon, Cloves, Coriander, Cumin, Ginger, Nutmeg, Paprika, Saffron, and Turmeric, selected for their economic importance, medicinal value, and widespread culinary use across global cuisines. Images were collected from Bangladesh, capturing diverse lighting conditions, backgrounds, natural textures, packaging styles, and multiple angles to reflect real-world scenarios. Stored in JPG format at a standardized 512×512 pixel resolution, the dataset is organized into dedicated folders with 1000 images per spice, ensuring consistency and ease of use. This resource is designed to facilitate machine learning and computer vision applications, such as spice recognition, classification, and quality assessment, as well as AI-driven tools for agricultural analysis and trade monitoring. By providing a robust and diverse visual dataset, this collection aims to advance research and to assist innovation in the global spice industry.

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Specifications Table

Subject	Spice Cultivar Classification using Machine Learning
Specific subject area	Image
Type of data	In this study, we collected 11,000 high-quality images of spices, captured in various environments, including both indoor and outdoor settings, under different daylight and nighttime conditions. We focused on 11 key spice categories based on market demand and size: black pepper, cardamom, cinnamon, cloves, coriander, cumin, ginger, nutmeg, paprika, saffron, and turmeric. We systematically organized the Images into separate folders for each spice category. We collected different types of images from diverse devices, angles, and lighting conditions, ensuring a rich and varied collection. To maintain image quality, we ensured a heterogeneous environment for data collection. We accumulated images in JPG format and resized them to a uniform resolution of 512×512 pixels for efficient storage and processing. We sequentially renamed the images to ensure a comprehensive organization within the dataset.
Data collection	
Data source location	Bangladesh
Data accessibility	Repository name: SpiceSpectrum: A Dataset of Spice Images Data identification number: 10.17632/5v7w2hx8n5.2 Direct URL to data: https://data.mendeley.com/datasets/5v7w2hx8n5/2

1. Value of the Data

Focused and Diverse Dataset: This dataset comprises 11 carefully selected spice categories, chosen based on current market demand and price trends. Each spice type is represented through a wide range of high-resolution images taken from multiple angles and under varying lighting conditions. This ensures diversity and richness in the dataset, offering a level of detail and variety not commonly available in existing spice image collections.

- **First Market-Driven, Open-Access, and Distinctive Dataset:** To the best of our knowledge, this is the first openly accessible dataset that focuses specifically on economically significant and high-demand spices. It uniquely combines visual data with insights derived from market analysis and economic trends, making it valuable across multiple research and application domains.
- **Designed for Machine Learning and Image Recognition:** This high-quality and diverse image dataset is optimized for training and evaluating computer vision models. The inclusion of varied capture settings enhances model robustness for real-world spice recognition, classification, and predictive analytics tasks. It serves as a strong benchmark resource for AI researchers.
- **Academic and Multi-disciplinary Applications:** This dataset offers significant value to academic research and applied fields alike. Potential applications include: Computer Vision and AI model development, Precision agriculture and post-harvest processing, Agricultural economics and market analysis, Mobile and embedded applications for real-time spice identification.
- **Practical Use in the Food and Agricultural Industry:** The dataset supports a broad range of food-tech innovations by enabling: automated quality control systems, smart inventory tracking for spice traders, and digital cataloging for e-commerce platforms. It also aids chefs, culinary professionals, food bloggers, and kitchen technology startups in building smarter food identification and tracking tools.
- **FAIR-Compliant Organization and Consistent Image Quality:** All images are captured under diverse real-world conditions but maintained at a standardized resolution, with a consistent and logical naming convention. This ensures the dataset is easily Findable, Accessible, Interoperable, and Reusable (FAIR), streamlining integration into machine learning workflows and supporting long-term reproducibility in research.

2. Background

While substantial research has been conducted on crops [1], vegetables [2], and fruits [3], the spice domain remains significantly underrepresented in publicly available datasets, especially those based on global market demand and trade relevance. Spices are not only vital culinary ingredients but also play an essential role in health, cosmetics, and international trade.

To address this gap, we introduce a market-oriented image dataset featuring 11 of the most economically significant spices worldwide: Black Pepper, Cardamom, Cinnamon, Cloves, Coriander, Cumin, Ginger, Nutmeg, Paprika, Saffron, and Turmeric. These were selected based on global demand, market pricing, medicinal value, and widespread use across various cuisines—from Indian and Middle Eastern to Latin American and Southeast Asian. Many of these spices, such as saffron, turmeric, and ginger, are extensively studied for their medicinal and pharmacological properties, commonly used in Sments, teas, and traditional healing [4]. Economically, they are key commodities in global trade, with major producers including India, Iran, Indonesia, and Spain. However, most existing spice datasets are limited in scope—often focusing solely on Indian varieties, lacking diversity in image backgrounds, and failing to reflect real-world market conditions such as lighting variations, packaging differences, or spice quality. Although the recently published Indian Spice Image dataset [5] contains images of spices, our dataset is a significantly better collection of spice images, as evidenced in Table 1.

The Indian Spice Image dataset [5] suffers from low image diversity, as visually depicted in Fig. 1. We noticed a lack of diversity in the Indian Spice Image dataset, therefore we aimed at capturing images of spices in various backgrounds and orientations.

To statistically prove the superior image diversity of our entire dataset, we have employed the following metrics:

Learned Perceptual Image Patch Similarity (LPIPS): LPIPS evaluates the perceptual similarity between two images [6]. Unlike traditional metrics like Mean Squared Error (MSE), which compare pixel values directly, LPIPS uses the internal activations of a deep neural network (VGG in this case) to mimic human perception of similarity. A high LPIPS score between two images means they are perceptually different. We calculated the average LPIPS score across many randomly sampled pairs of images within a dataset. A higher average LPIPS score indicates greater intra-class diversity. This means the images within that category exhibit a wider variety of appearances, such as different lighting, angles, backgrounds, and arrangements. It is observed from Table 2 that our dataset achieves better LPIPS score than the Indian Spice Images dataset on all classes of spices. For example, The LPIPS scores for SpiceSpectrum are consistently higher (e.g., 0.7494 vs. 0.5869 for Black Pepper), meaning its images are more perceptually distinct from one another.

Table 1
Comparison between our dataset and a previous dataset in terms of number of high market value spices.

Spice Names	Number of Images In SpiceSpectrum	Number of Images in Indian Spices Image Dataset [5]
Black Pepper	1000	278
Cardamom	1000	345
Cinnamon	1000	534
Cloves	1000	827
Coriander	1000	330
Cumin	1000	360
Ginger	1000	714
Nutmeg	1000	332
Paprika	1000	None
Saffron	1000	None
Turmeric	1000	None
Total	11,000	3360

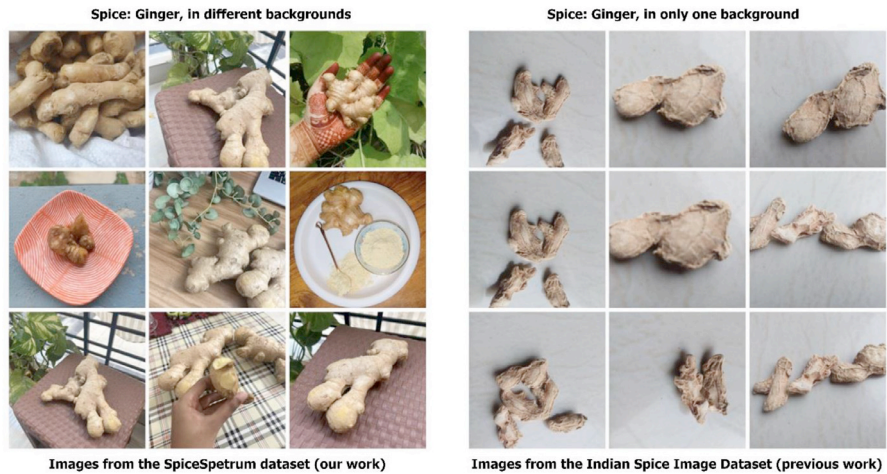


Fig. 1. Comparison of image diversity between our dataset (left) and a previous dataset (right) [5], indicating the Indian dataset lacks background and lighting diversity.

Table 2
Comparison of image diversity metric scores between our dataset and the Indian Spice Image Dataset.

	Diversity (LPIPS) ↑		Feature Volume (Trace) ↑		Quality (BRISQUE) ↓	
	Indian Spice Image Dataset	Spice Spectrum	Indian Spice Image Dataset	Spice Spectrum	Indian Spice Image Dataset	Spice Spectrum
Black pepper	0.5869	0.7494	101.04	319.73	37.68	26.45
Cardamom	0.4731	0.7358	105.87	309.04	41.92	27.95
Cinnamon	0.5527	0.7100	114.27	255.59	56.50	24.08
Cloves	0.6246	0.7468	156.96	334.58	34.24	30.21
Coriander	0.5234	0.7653	73.26	320.11	27.16	27.07
Cumin	0.5315	0.7204	67.46	282.65	27.15	22.37
Ginger	0.5222	0.7252	101.54	256.94	27.34	22.10
Nutmeg	0.4928	0.7407	98.52	248.42	47.91	26.68
Paprika	–	0.7593	–	276.17	–	20.95
Saffron	–	0.7117	–	308.92	–	25.52
Turmeric	–	0.7909	–	274.11	–	23.24

↑ indicates a higher score of the metric signifies better diversity; whereas ↓ implies a lower score means superior diversity.

Feature Volume (Trace of Covariance): Trace of the covariance matrix is a plausible estimate for image diversity measurement in current literature [7]. we first passed each image through a pre-trained neural network (ResNet-50) to extract a high-dimensional feature vector that represents its content. The “Feature Volume” is calculated as the trace of the covariance matrix of these feature vectors. The trace of the covariance matrix sums the variances along each feature dimension. A higher trace value signifies that the features are more spread out in the feature space. This implies the dataset contains a richer and more varied set of visual patterns (textures, shapes, colors, etc.). It complements LPIPS by measuring the overall “volume” occupied by the dataset’s features. It is clear from Table 2 that the Feature Volume scores for SpiceSpectrum are dramatically larger (e.g., 320.11 vs. 73.26 for Coriander), indicating a much wider spread of visual features. This suggests SpiceSpectrum contains spices photographed under more varied conditions.

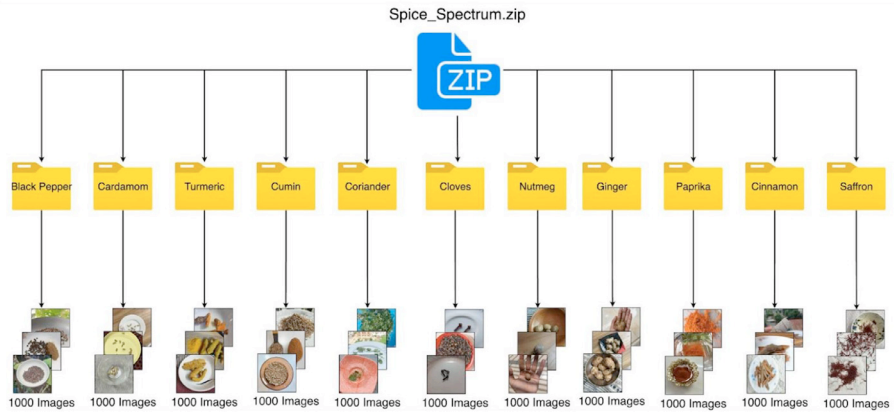


Fig. 2. Folder organization of our SpiceSpectrum dataset.

Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE): BRISQUE is a “no-reference” image quality assessment (IQA) model [8]. This means it can predict the perceptual quality of an image without needing a “perfect” or “pristine” version for comparison. It works by analyzing the image’s pixel statistics and comparing them to the statistics of natural, high-quality images. It outputs a score that correlates with human judgments of quality. A lower BRISQUE score indicates higher image quality, meaning fewer undesirable artifacts like blur, noise, compression artifacts, or unnatural distortions. A higher score suggests the image is of lower quality. The BRISQUE scores for SpiceSpectrum are consistently lower than for Indian Spice Image Dataset (e.g., 24.08 vs. a very high 56.50 for Cinnamon) in Table 2. This implies that images in Indian Spice Image Dataset are more likely to suffer from blur, noise, or other distortions, while images in SpiceSpectrum are perceptually cleaner and more “natural”.

To overcome these limitations, our dataset was collected from various environments in Bangladesh, an active participant in South Asia’s spice trade. The images capture real-world scenarios, including diverse lighting, natural textures, packaging styles, and multiple angles, as shown in Fig. 1. This makes the dataset highly suitable for training machine learning models and developing AI-powered tools for agricultural analysis, trade monitoring, and culinary research.

3. Data Description

This dataset comprises a comprehensive collection of 11,000 high-quality images representing 11 globally significant spices, curated to reflect the diversity of the global spice market. Each spice category: Black Pepper, Cardamom, Cinnamon, Cloves, Coriander, Cumin, Ginger, Nutmeg, Paprika, Saffron, and Turmeric are organized into their own dedicated folder, with 1000 images per category. The dataset is class-balanced. As shown in Fig. 2, the images were captured under various lighting conditions and from multiple angles to ensure consistency and diversity, making the dataset well-suited for a variety of computer vision and AI-based applications.

All images are stored in JPG format, a widely adopted standard that balances image quality with storage efficiency [9]. The resolution of each image is standardized at 512×512 pixels, ensuring clear and detailed visuals suitable for machine learning, visual recognition, and classification tasks. Within each category folder, images have been sequentially renamed to maintain an organized and systematic structure. Some samples from our dataset are shown in Fig. 3–5.



Fig. 3. Images of Nutmeg from our SpiceSpectrum dataset. Images were captured in different pots, natural backgrounds, in-hand, and various lighting conditions.



Fig. 4. Paprika images from our SpiceSpectrum dataset. Diverse images of this spice were collected to capture how it looks in different backgrounds.

4. Experimental Design, Materials and Methods

4.1. Global market analysis

The dataset was acquired through a systematic data collection process involving the capture and storage of high-quality spice images. We conducted a study on the global market on spices and found out the spices with the highest global market value: Black Pepper, Cardamom, Cinnamon, Cloves, Coriander, Cumin, Ginger, Nutmeg, Paprika, Saffron, and Turmeric. We selected the most economically valuable 11 spice varieties according to marked value and decided to collect images of these 11 spices. The global market value of the aforementioned spices are shown in [Table 3](#).



Fig. 5. Saffron images from our SpiceSpectrum dataset. Although Saffron is an expensive spice, we included diverse pictures of it in our dataset as it is an integral part of global cuisine.

Table 3

Approximate global market values of 11 widely traded spices selected for the dataset. These spices were prioritized for their economic significance and global demand, supporting the relevance of the collected visual data.

Spice	Approximate Global Market Value (USD)
Black Pepper	5.5 Billion [10]
Cardamom	64,580 Million [11]
Cinnamon	0.99 Billion [12]
Cloves	4.1 Billion [13]
Coriander	4.51 Billion [14]
Cumin	1.1 Billion [15]
Ginger	4.01 Billion [16]
Nutmeg	1.42 Billion [17]
Paprika	543.11 Million [18]
Saffron	567.19 Million [19]
Turmeric	4419.3 Million [20]

4.2. Data collection procedure

The collection procedure was designed to ensure diversity in image representation while maintaining consistency in quality and format. Five data annotators were assigned to collect the images from local markets in Bangladesh, capturing real-world scenarios such as varying lighting conditions, natural textures, orientations, packaging styles, backgrounds, and multiple angles. We took photos of multiple spices of the same variety to increase the diversity of our dataset. Each image was taken under diverse settings where possible, with some images captured in natural daylight and others in indoor lighting to reflect real-world market and kitchen conditions. The data collection process was structured into five distinct tasks (Task 1 to Task 5) for a single category of spice, as shown in Fig. 6. Each task required the collection of 200 photos per spice, totalling 1000 images per spice across the 11 spices, resulting in a dataset of 11,000 images. The workflow involved parallel data collection by the five data collectors (Data collector 1 to Data collector 5), with each data collector independently collecting 200 photos per spice. The processed images were then uploaded and stored in a centralized Google Drive folder for seamless access and management, as depicted in Fig. 6.

The images were captured by the data collectors using high-resolution smartphone cameras to ensure accessibility and variability in image quality and perspective. The specifications of the

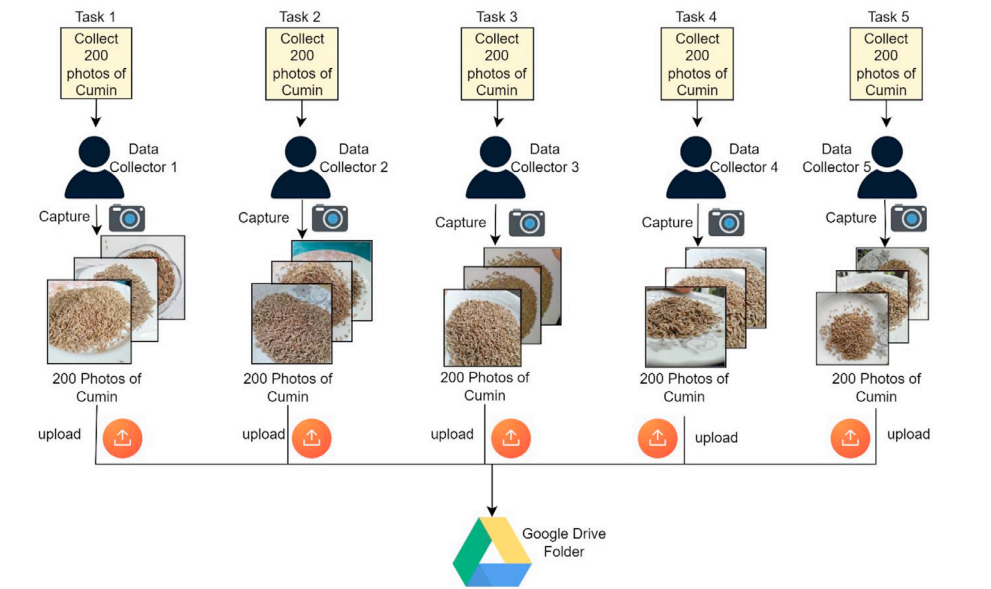


Fig. 6. Data collection process we followed to collect 1000 images of Cumin. Data collection tasks were equally distributed between 5 data annotators to ensure data diversity.

Table 4

Camera specifications of smartphones used by data collectors to capture images for our dataset. The varied specification of smartphone cameras makes our dataset more diverse.

Smartphone Model	Main Sensor	Lens & Optics	Focus & Stabilization
Poco M3	48 MP, 1/2.0" sensor size, 0.8 μm pixel size	f/1.79 aperture, 26 mm equiv. focal length	Phase Detection Autofocus (PDAF)
Redmi Note 9 Pro	64 MP, 1/1.72" sensor size, 0.8 μm pixel size	f/1.9 aperture, 26 mm equiv. focal length	Phase Detection Autofocus (PDAF)
iPhone 14	12 MP, 1/1.7" sensor size, 1.9 μm pixel size	f/1.5 aperture, 26 mm equiv. focal length	Dual Pixel PDAF, Sensor-shift Optical Image Stabilization (OIS)
Poco F1	12 MP, 1/2.55" sensor size, 1.4 μm pixel size	f/1.9 aperture	Dual Pixel PDAF
Google Pixel 8 Pro	50 MP, 1/1.31" sensor size, 1.2 μm pixel size	f/1.7 aperture, 25 mm equiv. focal length	Multi-zone Laser Detect AF (LDAF), All-pixel PDAF, Optical Image Stabilization (OIS)

smartphones used to capture the photos in our dataset are outlined in Table 4. The devices vary in terms of image resolution, sensor size, pixel size, aperture, focal length, focus, and stabilization features, making the images in our dataset different from one another.

The images were then processed using a Python (version 3.9) script with the Pillow library to resize them to a uniform resolution of 512 × 512 pixels, renaming them sequentially within each spice category folder (e.g., black_pepper_1.jpg to BlackPepper_1000.jpg), and convert all images to JPG format, ensuring consistency and efficient storage for machine learning applications. This procedure ensured that all annotated images were uniformly archived and easily accessible for future use. The entire methodology for developing the dataset is outlined in Fig. 7.

Sampling: A stratified sampling strategy was employed to curate a balanced dataset of 11,000 images across 11 spice categories. Images were systematically captured by five annotators from

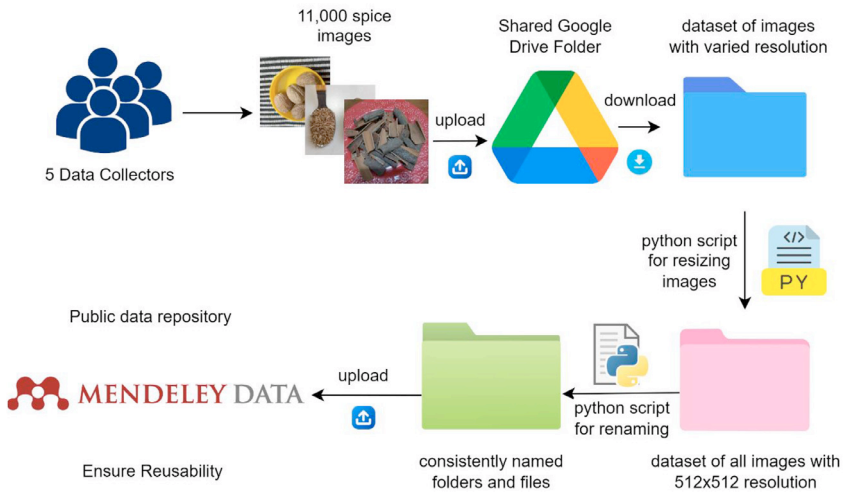


Fig. 7. Complete dataset creation methodology, including data collection and post-processing procedure.

local markets under diverse environmental conditions, including varied lighting, angles, and backgrounds to ensure authentic representation of real-world variability. The parallelized collection approach, with equal contributions per annotator and category, minimized bias and maximized intra-class diversity, thereby enhancing the dataset's robustness for downstream computer vision tasks.

Annotation Standards: Since images were directly uploaded by annotators into predefined category-specific Google Drive folders, manual annotation was unnecessary, ensuring inherent label accuracy. Clear guidelines were provided to annotators to maintain consistency during image collection. Regular cross-validation among annotators further minimized discrepancies, and only confidently identified images were retained. Metadata on capture conditions was preserved to support transparency and facilitate future dataset validation or extension.

Preprocessing Code and Metadata Availability: All images were uniformly resized, converted, and renamed using a Python script with the Pillow library. Accompanying metadata, including camera specifications and capture conditions, are embedded in the image files to support reproducibility. The full preprocessing code and metadata are openly accessible at <https://github.com/noushad999/SpiceSpectrum>

Limitations

All of the images contained in the SpiceSpectrum dataset were collected from Bangladesh, potentially limiting the dataset's representativeness to other regions of the world. Hence, we tried our best to capture images in diverse environments to increase representativeness as much as possible. Even though we collected images of 11 spice varieties, some commercially valuable spices such as Vanilla, Star Anise, Fenugreek, Mustard Seeds, Allspice, etc. are not included in our dataset, indicating future scope of further enhancing our dataset.

Ethics Statement

The authors have read and follow the ethical requirements for publication in Data in Brief and confirm that the current work does not involve human subjects, animal experiments, or any data collected from social media platforms.

CRediT Author Statement

Md. Noushad Jahan Ramim: Writing - Original Draft, Data Curation, Visualization, Software. **Samira Islam:** Data Curation, Resources, Writing - Review & Editing. **Muhtasin Towkir:** Data Curation, Visualization. **Md Muhtasim Fuad:** Data Curation, Supervision. **Noor Mairukh Khan Arnob:** Data Curation, Project administration, Software.

Data Availability

SpiceSpectrum: A Dataset of Spice Images (Original data) (Mendeley Data)

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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